

A First Course in Causal Inference Ch.4–5

Kodai Minesaki

Faculty of Engineering, The University of Tokyo

August 22, 2026

Ch.4 Neymanian Repeated Sampling Inference in CRE

Ch.4 Neymanian Repeated Sampling Inference in CRE

- ▶ Neyman (1923) has proposed an unbiased point estimator and a conservative confidence interval based on the sampling distribution of the point estimator
- ▶ This chapter covers Neyman (1923)'s fundamental parts.

4.1 Finite population quantities

- ▶ Setup: CRE with n units (treatment group: n_1 units, control group: n_0 units); For $i = 1, \dots, n$, potential outcomes $Y_i(1)$ and $Y_i(0)$ and individual effect $\tau_i = Y_i(1) - Y_i(0)$.
- ▶ We then obtain finite population means:

$$\bar{Y}(1) = \frac{1}{n} \sum_{i=1}^n Y_i(1), \quad \bar{Y}(0) = \frac{1}{n} \sum_{i=1}^n Y_i(0),$$

- ▶ variances:

$$S^2(1) = \frac{1}{n-1} \sum_{i=1}^n \{Y_i(1) - \bar{Y}(1)\}^2, \quad S^2(0) = \frac{1}{n-1} \sum_{i=1}^n \{Y_i(0) - \bar{Y}(0)\}^2,$$

- ▶ and covariance:

$$S(1, 0) = \frac{1}{n-1} \sum_{i=1}^n \{Y_i(1) - \bar{Y}(1)\} \{Y_i(0) - \bar{Y}(0)\}.$$

4.1 Finite population quantities

- ▶ The individual effects have mean

$$\tau = \frac{1}{n} \sum_{i=1}^n \tau_i = \bar{Y}(1) - \bar{Y}(0),$$

- ▶ and variance

$$S^2(\tau) = \frac{1}{n-1} \sum_{i=1}^n (\tau_i - \tau)^2$$

We then have

Lemma 4.1

$$2S(1, 0) = S^2(1) + S^2(0) - S^2(\tau).$$

4.2 Neyman (1923)'s theorem

- ▶ We are interested in estimating the average causal effect τ based on the (observed) data $(Z_i, Y_i)_{i=1}^n$ from a CRE.
- ▶ Based on the observed data, the sample means:

$$\hat{\bar{Y}}(1) = \frac{1}{n} \sum_{i=1}^n Z_i Y_i, \quad \hat{\bar{Y}}(0) = \frac{1}{n} \sum_{i=1}^n (1 - Z_i) Y_i,$$

- ▶ and the sample variances:

$$\hat{S}^2(1) = \frac{1}{n_1 - 1} \sum_{i=1}^n Z_i \{Y_i - \hat{\bar{Y}}(1)\}^2, \quad \hat{S}^2(0) = \frac{1}{n_0 - 1} \sum_{i=1}^n (1 - Z_i) \{Y_i - \hat{\bar{Y}}(0)\}^2.$$

- ▶ Obviously, there are no sample covariance and $S^2(\tau)$ because **we cannot observe both potential outcomes for each unit**.

4.2 Neyman (1923)'s theorem

Theorem 4.1

1. the difference-in-means estimator $\hat{\tau} = \hat{\bar{Y}}(1) - \hat{\bar{Y}}(0)$ is unbiased for τ :

$$\mathbb{E}[\hat{\tau}] = \tau.$$

2. $\hat{\tau}$ has variance

$$\text{var}(\hat{\tau}) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n} = \frac{n_0}{n_1 n} S^2(1) + \frac{n_1}{n_0 n} S^2(0) + \frac{2}{n} S(1, 0).$$

3. the variance estimator $\hat{V} = \frac{\hat{S}^2(1)}{n_1} + \frac{\hat{S}^2(0)}{n_0}$ is conservative for estimating $\text{var}(\hat{\tau})$ in the sense that $\mathbb{E}[\hat{V}] - \text{var}(\hat{\tau}) = \frac{S^2(\tau)}{n} \geq 0$ with equality holding if and only if $\tau_i = \tau$ for all units.

4.2 Neyman (1923)'s theorem

- ▶ The FRT works for any test statistic but Neyman (1923)'s theorem is only about the difference in means.
- ▶ If $S(1, 0) > 0$, then those treated units also have relatively large control potential outcomes¹. $\rightsquigarrow$ the observed outcomes under control are relatively small $\rightsquigarrow$ large $\hat{\tau}$.
- ▶ If $S(1, 0) < 0$, the those treated units have relatively small control potential outcomes². $\rightsquigarrow$ the observed outcomes under control are relatively large $\rightsquigarrow$ small $\hat{\tau}$.

¹In other words, large control potential outcomes are not “observed.”

²In other words, small control potential outcomes are not “observed.”

4.2 Neyman (1923)'s theorem

Thorem 4.2

Let $n \rightarrow \infty$ and $n_i \rightarrow \infty$. If $\frac{n_1}{n}$ has a limiting value in $(0, 1)$, $\{S^2(1), S^2(0), S(1, 0)\}$ have limiting values, and

$$\max_{1 \leq i \leq n} \frac{\{Y_i(1) - \bar{Y}(1)\}^2}{n} \rightarrow 0, \quad \max_{1 \leq i \leq n} \frac{\{Y_i(1) - \bar{Y}(0)\}^2}{n} \rightarrow 0, \quad (1)$$

then

$$\frac{\hat{\tau} - \tau}{\sqrt{\text{var}(\hat{\tau})}} \xrightarrow{d} \mathcal{N}(0, 1), \quad (2)$$

and

$$\hat{S}^2(1) \xrightarrow{p} S^2(1), \quad \hat{S}^2(1) \xrightarrow{p} S^2(0) \quad (3)$$

4.2 Neyman (1923)'s theorem

The intuitive understanding for Thm. 4.2 is:

- ▶ Eq. 1: potential outcomes for each unit are not extremely large to dominate the variability³.
- ▶ Suppose a case one unit has an extremely large outcome, then $\hat{\tau}$ depends on which group the unit belongs to: Eq. 2 does not hold.
- ▶ Eq. 3 ensures Neyman (1923)'s conservative CI for τ , because $\hat{V} \geq \text{var}(\hat{\tau})$.

³Formally, $\max_{1 \leq i \leq n} \|Y_i(1) - \bar{Y}(1)\| = o(\sqrt{n})$.

4.3 Proofs

Unbiasedness of $\hat{\tau}$:

$$\hat{\tau} = \frac{1}{n_1} \sum_{i=1}^n Z_i Y_i - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) Y_i = \frac{1}{n_1} \sum_{i=1}^n Z_i Y_{i(1)} - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) Y_{i(0)},$$

then

$$\begin{aligned} \mathbb{E}[\hat{\tau}] &= \mathbb{E} \left[\frac{1}{n_1} \sum_{i=1}^n Z_i Y_{i(1)} - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) Y_{i(0)} \right] \\ &= \frac{1}{n_1} \sum_{i=1}^n \mathbb{E}[Z_i] Y_{i(1)} - \frac{1}{n_0} \sum_{i=1}^n \mathbb{E}[1 - Z_i] Y_{i(0)} \\ &= \frac{1}{n_1} \sum_{i=1}^n \frac{n_1}{n} Y_{i(1)} - \frac{1}{n_0} \sum_{i=1}^n \frac{n_0}{n} Y_{i(0)} = \frac{1}{n} \sum_{i=1}^n Y_{i(1)} - \frac{1}{n} \sum_{i=1}^n Y_{i(0)} = \tau. \quad \square \end{aligned}$$

4.3 Proofs

$\text{var}(\hat{\tau})$: We write $\hat{\tau}$ as

$$\hat{\tau} = \sum_{i=1}^n Z_i \left\{ \frac{Y_i(1)}{n_1} + \frac{Y_i(0)}{n_0} \right\} - \frac{1}{n_0} \sum_{i=1}^n Y_i(0).$$

Then, the variance of $\hat{\tau}$ follows:

$$\begin{aligned} \text{var}(\hat{\tau}) &= \frac{n_1 n_0}{n(n-1)} \sum_{i=1}^n \left\{ \frac{Y_i(1)}{n_1} + \frac{Y_i(0)}{n_0} - \frac{\bar{Y}(1)}{n_1} - \frac{\bar{Y}(0)}{n_0} \right\}^2 \\ &= \frac{n_1 n_0}{n(n-1)} \left[\frac{1}{n_1^2} \sum_{i=1}^n \{Y_i(1) - \bar{Y}(1)\}^2 + \frac{1}{n_0^2} \sum_{i=1}^n \{Y_i(0) - \bar{Y}(0)\}^2 \right. \\ &\quad \left. + \frac{2}{n_1 n_0} \sum_{i=1}^n \{Y_i(1) - \bar{Y}(1)\} \{Y_i(0) - \bar{Y}(0)\} \right] \\ &= \frac{n_0}{n_1 n} S^2(1) + \frac{n_1}{n_0 n} S^2(0) + \frac{2}{n} S(1, 0). \end{aligned}$$

4.3 Proofs

Here, the first equality holds because of the following Lemma:

Lemma C.2

Under simple random sampling, the sample means are unbiased for the population means:

$$\mathbb{E} \left[\widehat{\bar{c}} \right] = \bar{c}, \quad \mathbb{E} \left[\widehat{\bar{d}} \right] = \bar{d}.$$

Their variances and covariance are

$$\text{var} \left(\widehat{\bar{c}} \right) = \frac{n_0}{nn_1} S_c^2, \quad \text{var} \left(\widehat{\bar{d}} \right) = \frac{n_0}{nn_1} S_d^2, \quad \text{cov} \left(\widehat{\bar{c}}, \widehat{\bar{d}} \right) = \frac{n_0}{nn_1} S_c d.$$

4.3 Proofs

Let $A_i = \frac{Y_i(1)}{n_1} + \frac{Y_i(0)}{n_0}$, then $\text{var}(\hat{\tau}) = \text{var}(\sum_{i=1}^n Z_i A_i)$. In a CRE, we draw a simple random sample of size n_1 for a treatment group. Then, $\bar{A} = \frac{1}{n_1} \sum_{i=1}^n Z_i A_i$. Following Lemma C.2, $\text{var}(\bar{A}) = \frac{n_0}{nn_1} S_A^2$. Hence,

$$\text{var}(\hat{\tau}) = \text{var}\left(\sum_{i=1}^n Z_i A_i\right) = \text{var}(n_1 \bar{A}) = n_1^2 \text{var}(\bar{A}) = n_1^2 \frac{n_0}{nn_1} S_A^2 = \frac{n_0 n_1}{n} S_A^2.$$

Because $S_A^2 = \frac{1}{n-1} \sum_{i=1}^n \{A_i - \bar{A}\}^2$, we obtain

$$\text{var}(\hat{\tau}) = \frac{n_0 n_1}{n(n-1)} \sum_{i=1}^n \left\{ \frac{Y_i(1)}{n_1} + \frac{Y_i(0)}{n_0} - \frac{\bar{Y}(1)}{n_1} - \frac{\bar{Y}(0)}{n_0} \right\}^2$$

4.3 Proofs

From Lemma 4.1, we also write the variance as

$$\hat{\tau} = \frac{n_0}{n_1 n} S^2(1) + \frac{n_1}{n_0 n} S^2(0) + \frac{1}{n} (S^2(1) + S^2(0) - S^2(\tau)) = \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}. \quad \square$$

4.4 Regression analysis of the CRE

- ▶ Practitioners often use regression-based inference for the average causal effect τ .
- ▶ A standard approach is OLS:

$$(\hat{\alpha}, \hat{\beta}) = \arg \min_{(a,b)} \sum_{i=1}^n (Y_i - a - bZ_i)^2.$$

- ▶ Here, $\hat{\tau} = \hat{\beta}$.
- ▶ However, the usual variance estimator from the OLS equals

$$\hat{V}_{\text{OLS}} = \frac{n(n_1 - 1)}{(n - 2)n_1 n_0} \hat{S}^2(1) + \frac{n(n_0 - 1)}{(n - 2)n_1 n_0} \hat{S}^2(0) \approx \frac{\hat{S}^2(1)}{n_0} + \frac{\hat{S}^2(0)}{n_1},$$

where the approximation holds with large n_1 and n_0 (it can differ).

- ▶ EHW robust variance estimator is closer to $\hat{V}$ and HC2 variant of the EHW robust variance estimator is identical to $\hat{V}$.

Ch.5 Stratification and Post-Stratification in Randomized Experiments

Ch.5 Stratification and Post-Stratification in Randomized Experiments

This chapter explains the meaning of the following quote:

Block what you can and randomize what you cannot.

– *Box et al. (1978)*

- Box, G. E., W. H. Hunter, S. Hunter, et al. (1978). *Statistics for experimenters*, Volume 664. John Wiley and sons New York.
- Neyman, J. (1923). On the application of probability theory to agricultural experiments. essay on principles. *Ann. Agricultural Sciences*, 1–51.